🧩 Assignment Title:

Design and Prototype of a Smart Industrial Tagging Application

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🎯 Objective

Students will design a **mobile application prototype using Figma** that demonstrates how **tagging technologies** (RFID, NFC, or QR/barcode) can be used to improve industrial operations — such as asset tracking, inventory management, worker safety, or maintenance scheduling.

The goal is to understand the role of **ubiquitous identification (tagging)** in pervasive computing environments and how it integrates with mobile systems and backend services.

📘 Assignment Description

Each group of students will:

1. Select an **industrial context** (e.g., factory, warehouse, hospital, port, cold storage, etc.).
2. Identify **assets, tools, or people** that will be tagged using **RFID/NFC/Barcode**.
3. Design a **mobile application prototype in Figma** showing how users (technicians, managers, or logistics operators) interact with these tags.
4. Describe how **data flows** from the tagged item to the application (through scanning, reading, or auto-detection).
5. Present the system concept and app prototype in class.

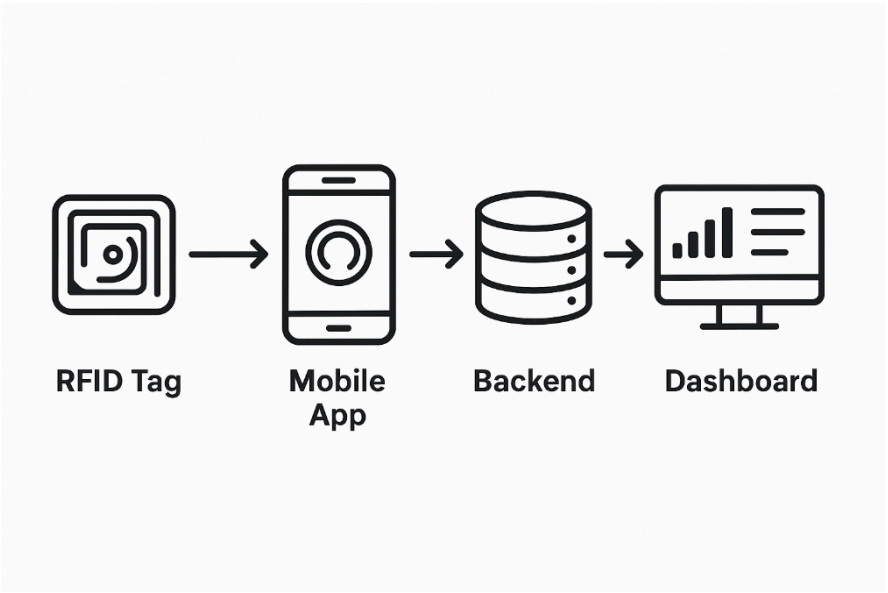


Figure 1 Flow of Application

🧠 Guidelines for Application Design

Your Figma design **must include the following screens and components**:

Screen	Required Elements	Notes / Example Use
1️⃣ **Splash & Login**	Logo, RFID/NFC/QR login or manual login	Optional: Show company name, loading animation
2️⃣ **Dashboard / Home**	Summary cards (e.g., "Total Assets," "Active Tags," "Pending Maintenance")	Use icons, minimal layout
3️⃣ **Tag Scanning Screen**	Tag Reader interface — simulate **RFID/NFC scan** or **camera-based barcode scan**	Show scanned item info dynamically
4️⃣ **Asset / Item Detail**	Tag ID, Name, Location, Condition, Last Updated, History	Add edit/update button
5️⃣ **Tag Assignment / Registration**	Form to assign tag to new item or equipment	Include fields for Tag ID, Asset name, category
6️⃣ **Notification & Alerts**	List of system alerts or maintenance reminders	Example: "Machine A overdue maintenance"
7️⃣ **Analytics / Report (Optional)**	Visualization of usage, maintenance, or movement history	Charts or bar graph (optional but adds value)

🧩 Functional Concept (What You Must Explain)

In your **documentation or presentation**, include:

1. **Tagging technology used** (RFID / NFC / QR)
   - Example: RFID passive tag attached to tools for tracking.
2. **Data flow diagram (DFD / system flow)**
   - How data moves from tag → reader → mobile app → database.
3. **User roles and interaction**
   - Example: Operator scans tag; Manager reviews status via app.
4. **Pervasive computing element**
   - How the system operates "in the background" or provides **context-aware** information.
5. **Integration possibilities**
   - Example: Integration with IoT sensors, backend cloud (e.g., Firebase), or ERP.

💡 Example Project Ideas

Scenario	Tagging Use Case	Expected Mobile Feature
Factory tool tracking	Each tool has RFID tag	Tag scan + Maintenance history
Hospital patient tracking	NFC wristband for patients	Tag-based identity + room tracking
Warehouse inventory	QR label on boxes	Batch scan + Quantity update

Scenario	Tagging Use Case	Expected Mobile Feature
Cold storage monitoring	RFID tag + temperature sensor	Alert when out of range
Port container logistics	RFID on container	Tag scan + Movement record

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🧾 Submission Requirements

- **Deliverables:**
  1. Figma project link (shared publicly or view-only)
  2. One-page system concept document (PDF)
  3. Screenshot of app flow (exported from Figma)
- **Submission format:** Upload to LMS or send via email with subject line: `UbiComp_Assignment_[GroupName]_[TaggingType]`
- **Presentation:** 5–7 minutes demonstration per group.